Review 2

PROJ2999 Capstone Project Introduction

MENTAL HEALTH ANALYSIS USING MACHINE LEARNING MODELS

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Outline

1 Recap of Review 1

2 Problem Statement and Objectives

3 Requirement Analysis

4 Identification of tools/technology/algorithms

5 Proposed Methodology

6 Project Workflow

7 Data Flow or Use Case Diagram

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Feedback from Guide/Reviewers and Actions Taken after Review 1

Feedback Received in Review 1

- Expand the project beyond questionnaire-based analysis.
- Avoid relying on older models such as SVM and Logistic Regression; explore updated models from research papers.
- Include only information that the team can clearly explain and justify.

Action Taken

- Added psychological text data as an additional data source alongside questionnaire data.
- Selected DeBERTa for psychological text analysis and FT-Transformer for questionnaire analysis.
- Planned two separate models whose outputs will be combined to predict depression, anxiety and stress.
- Completed collection of both datasets.

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Aim of the Project

Aim

To investigate whether machine-learning models can use relevant demographic, behavioural, lifestyle, academic/workplace, social and psychological factors to support early screening for possible depression, anxiety and stress.

Problem Statement

Mental-health concerns may remain unrecognised until individuals report symptoms or receive professional evaluation. This project will investigate whether machine-learning models can combine demographic, behavioural, lifestyle, academic/workplace, social and psychological factors to identify individuals who may be at risk of depression, anxiety or stress. The framework will compare predictive models and explain contributing factors to support screening and professional decision-making.

Objectives

1. Analyse questionnaire data using the FT-Transformer model.
2. Analyse psychological text data using the DeBERTa model.
3. Combine the outputs of both models to generate predictions for depression, anxiety and stress.

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Requirement Analysis

- Use the benchmark questionnaire and psychological text datasets.
- Inspect data formats, missing values, labels and data quality.
- Preprocess structured questionnaire responses and psychological text.
- Analyse questionnaire data using FT-Transformer.
- Analyse psychological text using DeBERTa.
- Combine model outputs to predict depression, anxiety and stress.
- Provide clear and readable outputs.

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Identification of Tools/Technology/Algorithms

Tools

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

Algorithms

- FT-Transformer (Questionnaire data)
- DeBERTa (Psychological text)

The selected models are proposed for two separate data streams. Their performance will be assessed after preprocessing, training and testing.

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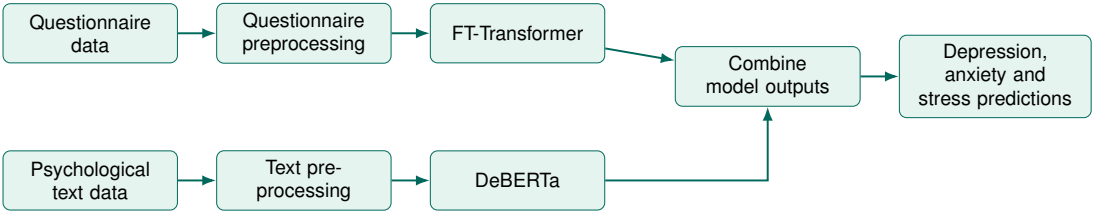
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Proposed Methodology



Methodology

The questionnaire and psychological text data will be processed through separate pipelines. FT-Transformer will analyse questionnaire data, while DeBERTa will analyse psychological text. The outputs of the two models will be combined to generate predictions for depression, anxiety and stress.

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Proposed Workflow

Questionnaire Pipeline

1. Collect questionnaire data.
2. Preprocess questionnaire responses.
3. Analyse the data using FT-Transformer.

Psychological Text Pipeline

1. Collect psychological text data.
2. Preprocess psychological text.
3. Analyse the text using DeBERTa.

Combined Output

1. Provide questionnaire and psychological text inputs from the same user during testing.
2. Combine the outputs of FT-Transformer and DeBERTa.
3. Generate predictions for depression, anxiety and stress.

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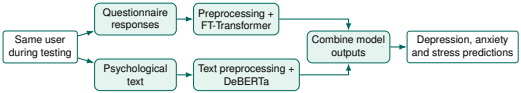
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Data Flow Diagram (Conceptual)



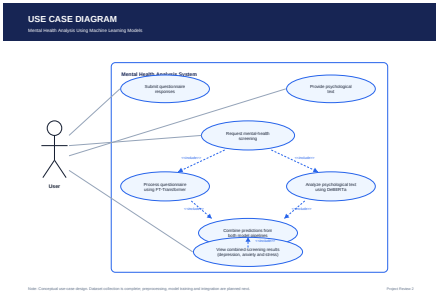


Figure: Use Case Diagram

Diagram Scope

During testing, questionnaire responses and psychological text will be provided by the same user. Each input will be processed by its respective model, and the model outputs will be combined to predict depression, anxiety and stress.

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Current Project Status

The project focuses on analysing questionnaire data and psychological text data using two separate models: FT-Transformer and DeBERTa. The outputs will be combined to generate predictions for depression, anxiety and stress.

Work Completed

- Collection of questionnaire dataset completed.
- Collection of psychological text dataset completed.
- Model selection completed: FT-Transformer and DeBERTa.
- Proposed methodology and workflow planned.

Next Step

Data preprocessing is pending and will be the next step, followed by model development, integration, testing and evaluation.

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Workplan and Proposed Timeline

Task	Task owner	Sept	Oct	Nov
Problem identification	Mrudula,Vivek	Done		
Literature survey	Mrudula	Done		
Project objectives	Priya	Done		
Workplan	Priya,Mrudula	Done		
Requirement analysis	Vivek,Rishi		Done	
Identification of Tools/algorithms/Algorithms	Vivek,Mrudula		Done	
System Design(DFD or UMLs)	Rishi		Plan	
Proposed methodology	Priya,Vivek			Plan
Closure of Phase1 / Project Exhibition	Team			Plan

Table: Proposed timeline of the project

References

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3. B. Bailey and D. R. Strunk, “Predicting anxiety and depression over 12 months of the COVID-19 pandemic: A machine learning study,” *Journal of Clinical Psychology*, vol. 79, pp. 2388–2403, 2023. doi: 10.1002/jclp.23555.
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5. P. He, X. Liu, J. Gao, and W. Chen, “DeBERTa: Decoding-enhanced BERT with Disentangled Attention,” arXiv:2006.03654, 2020. https://arxiv.org/abs/2006.03654.
6. “Revisiting Deep Learning Models for Tabular Data,” arXiv:2106.11959, 2021. https://arxiv.org/abs/2106.11959.

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Disclosure Statements

- I will maintain academic honesty by doing my own work and avoiding plagiarism, cheating, unauthorized collaboration, and falsifying information, use of AI Content.
- I will properly acknowledge and cite all sources I use, including information from the internet, books, software, and AI tools.
- I will use institutional resources responsibly and only for authorized academic or research activities. I understand that I am responsible for my own actions and decisions.